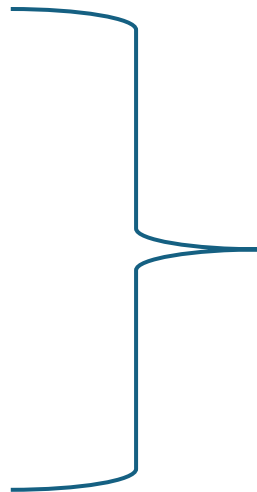


Data Analysis

Data analysis

- Digging deep into data that we have collected in the field / from an experiment.
 - Trends
 - Patterns
 - Explain a story from the data from
 - Behaviour
 - Practices
 - Culture
 - Lab experiments
 - System / user interaction
- 
- From the research

Approaches of data analysis

- Based research paradigms:
 - Ontology
 - Epistemology
- Positivists
 - Dominated by numeric data, to some extent, there can also subjective / text data
- Interpretivists
 - Dominated by subjective data / also have quantitative data

Quantitative data analysis

- Strategies
 - Descriptive data analysis
 - Basic statistical analysis
 - Techniques : tables, ratios, charts/graphs, averages and standard deviations
 - This is applicable when analysing single variable data. I,e, we do not seek to understand a relationship between two or more variables.
 - Correlation data analysis {inference data analysis}
 - Is used to analyse data with more than one variable where we see to establish a relationship between those variable.
 - Techniques include the T-Test and regression analysis

Qualitative data management

- The data may include: pictures, words, voices, videos etc.
- Approaches:
 - Deductive-based on pre defined concepts
 - Inductive- not constrained to predefined concepts
- Qualitative data is analysed using thematic data analysis
 - Establishing word, phrase, sentence that best describes pattern on a phenomenon under study
- Data analysis starts in the course of data collection
 - Reflections:
 - Take notes of observation

Stages in qualitative data analysis

- Familiarisation
 - Check all data available
 - Notes are clear
 - Transcribing : converting your data from notes format to complete readable text. Essay– Transcripts
- Coding: a word, or phrase of our interest in relation to the study
 - E.g. Security : coded by color, line etc..
- Categorisation : Grouping related code
- Themes: develop these from the categories

Sampling

Sample

- A sample is a representation of a study population.
- Study population refers to the expected WHOLE participants in research.
- We are expected to involved everyone in a study:
 - Its expensive
 - Its time consuming
 - Hence, research used a sample of the participants
- Whatever size, but characteristically, the sample must be a true reflection of the population

Guidelines for selecting a sample

- Must be one that gives you adequate information about your study
- Must not be selected with biasness, hence, you should be clear how you selected them, e.g.
 - Purposive sampling
 - Random sampling
- Must accessible
- In quantitative : use formula to determine size
- In qualitative : information saturation and or information richness

Assignment

Write a structured research concept on any research problem of your choice in ICT field. This should be a 1-page work, excluding the cover page and references page.

The work should be structured as follows:

Cover page

Assignment page

Background & Literature

Problem

Aim

Objectives

Study approach and research strategy

Data Collection

Data Analysis

Reference (Citation using APA 7th Edition)

Submission Guide:

- Date: 28th April 2026
- One soft copy in MS Word to (<https://drive.google.com/drive/folders/1JHPH0Ogz9Cqx0kxCWdj54DnUchDyB2P8?usp=sharing>)
- File name must
PROG_GROUP_NAME_NAME_OF_SENDER
E.G. *BISO_Group1_martinmsendema*
- One Hard Copy by 16:30Hrs
- Formatting: Arial 12, before 0, after 0, line spacing 1.5, justify
- Separate for groupd for BIT, BIS, (BITW-BISW),(BIT-BIS-ODEL)
 - 3 per group